CENTRALE NANTES

# Brochure

# CNFS 2025-2026

PREPARE THE FUTURE

# Formula Student




## The Objective

To train the engineers of tomorrow through complex, comprehensive industrial projects at the cutting edge of technology.

Formula Student isn't just a speed race; it's a comprehensive evaluation of a prototype built entirely by students over the course of a year, through two types of challenges:

- Dynamic Challenges: pure performance tests culminating in the 22-kilometer endurance race, the true test of reliability.
- Static Challenges: the defense of various engineering choices before industry experts (Design, Costs, Business Plan).

## +700 UNIVERSITIES

It is the world's largest student competition, having been held since 1981



# Formula Student France

Since 2023, France has joined Germany, the United Kingdom, Austria, and many others in hosting the competition on its own soil. And despite its relative youth, enthusiasm is sky-high: since 2023, the number of participating teams has tripled!

**+20**

FRENCH SCHOOL
registered in 2026

**8**

DIFFERENT COUNTRIES
involved in 2026

**+150**

STUDENT TEAMS
in 2026



# The Team






## SIMON POIRET : THE EXPERT

Researcher at LHEEA, Professor of Propulsion, Former FS Team Leader in Sweden... Without his expertise, this project wouldn't exist!



## THEO STIPA : THE PRESIDENT

The man behind the project—a seasoned pilot, a simulation enthusiast, a veritable walking encyclopedia of cars, and a CAD whiz—Théo is the team's ace!



## KARL KNUYT : THE VP

He knows the ins and outs of the organization and takes as much of the load off Théo as possible so the whole team can move faster! He ensures we meet our environmental and ethical commitments.



## ELWAN EL-HAFAIA : THE TREASURER

Since he doesn't have a well-stocked bank account just yet, Elwan handles design, communications, and the business plan—and dreams of making money.



## AXEL NAVARRETE : THE ELECTRONICS

A mechatronics engineer who came from Mexico to pursue a dual degree in Nantes. He handles electronics and electrical safety like a pro!



## PIERRE BACCINI : THE CONCEPTION

From the braking system to the pedal assembly and even chassis components, Pierre isn't afraid of any CAD work or calculations!



## SAKSHAM SAGAR : V.DYNAMIC

Saksham, a former member of the SAE HELIOS Racing team from India, brings his expertise in dynamic simulation to the team!



## AND MUCH MORE !

Many other students have already joined us because they believe in this project! Plus, CNFS is recruiting at all levels! Don't hesitate to contact us!

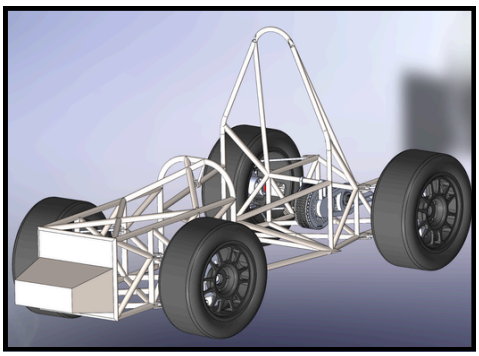
# Our objectives

## A word from Théo



**“Launching CNFS is a unique challenge. With the Falcon, we’re bringing Centrale Nantes’ excellence to the racetrack. Our goal: to prove that sustainable engineering goes hand in hand with performance. Together, we’re building the future of electric mobility and learning to be the engineers of tomorrow.”**

## 1. THE CONCEPT CLASS



The first mandatory step! Having our complete design and business plan thoroughly reviewed by an experienced and professional panel of judges. This provides precise, objective feedback.

## 2. THE CONCEPTION



With our digital model already well underway, we can start purchasing and manufacturing some of the parts we’ve designed! Once we’ve secured the necessary funding from partners and sponsors!

## 3. THE CHALLENGE



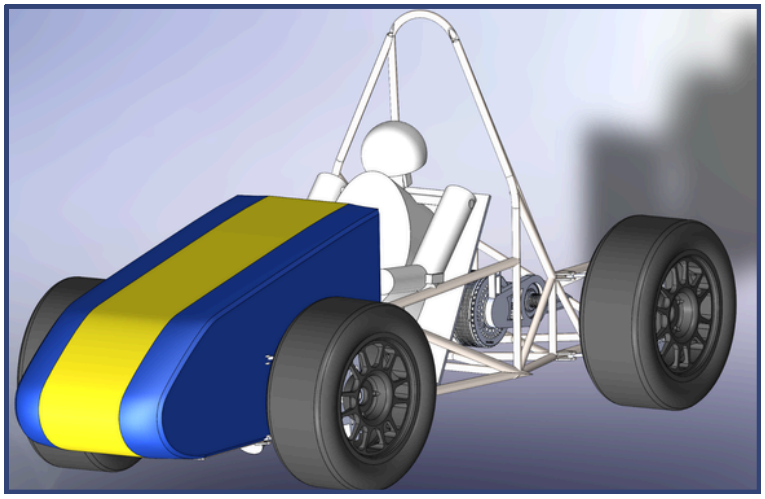

The ultimate goal! To compete with other universities across Europe! First goals: FS FRANCE and FS AUSTRIA in 2027!

# Our commiments

## ENVIRONMENTAL PROTECTION AT THE HEART OF OUR CONCERNS

In line with current environmental concerns, the CNFS team is competing in the electric vehicle category. We've swapped the internal combustion engine for an Emrax electric motor (see opposite), which allows us to reach speeds of up to 125 km/h!





## OUR GUIDING PRINCIPLES: SAFETY & PROACTIVITY

Two team members will compete as drivers in our prototype, which can accelerate from 0 to 100 km/h in under 3.5 seconds. It is therefore essential to have a thorough understanding of every aspect of the car. This involves months of computer-aided design and simulations under the most challenging conditions encountered in racing or during jury tests, all while adhering to the competition's 150-plus-page set of regulations. This has now led to a full-scale physical model (one iteration of which is shown here) of the first car: the Falcon

# The figures

**+50K€**

NEEDED for the manufacture of the car

**+210 KG**

We want to minimize the expected weight of the prototype

**10**

MONTHS to make the project a reality

**+110 KM/H**

Target speed of the prototype

**+15**

SUBSYSTEMS to be modeled and optimized

**+150**

STUDENT TEAMS listed in 2026

# How to contact us

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